

AI-Assisted Message Routing in Pub/Sub Systems

For 100 hours project

The students will design and implement a lightweight pub/sub system that integrates basic AI-assisted message routing. The goal is to classify messages based on priority levels (e.g., real-time vs. batch processing) and direct them to the appropriate subscribers. The system will use LLMs to analyze message content and suggest routing policies. Instead of developing a full-fledged broker, students will build on existing frameworks like MQTT, Kafka, or RabbitMQ. The project will focus on understanding pub/sub architectures, implementing simple LLM-driven classification, and evaluating latency and throughput.

For 200 hours project (additional tasks)

- Implement basic user-defined routing policies (e.g., filtering by topic).
- Support traffic balancing between multiple subscribers.
- Extend the system with fault tolerance mechanisms (e.g., retry logic for dropped messages).

Phase 1: Research & Understanding (10-15% of total time)

Goal: Familiarize the student with pub/sub messaging, existing message brokers (e.g., MQTT, Kafka, RabbitMQ), and AI-based routing approaches.^[1]_{SEP}**Tasks:**

- Study Pub/Sub architectures and message brokers.
- Explore existing routing mechanisms in pub/sub systems.
- Review basic AI techniques for message classification (e.g., keyword-based, rule-based AI, or LLM-driven classification).
- Identify which broker (MQTT, Kafka, or RabbitMQ) best suits the project.

Deliverable: Initial literature review + choice of pub/sub framework.

Phase 2: System Design & Specification (15-20%)

Goal: Define the architecture for AI-assisted message routing.^[1]_{SEP}**Tasks:**

- Define message types and routing criteria (e.g., priority-based, content-based).
- Design a lightweight AI-based routing mechanism (e.g., LLM-generated routing rules or rule-based AI filtering).
- Decide where AI will be integrated (broker-level vs. external filtering layer).
- Draft a simple architecture diagram and key system components.

Deliverable: System design document with architecture, routing rules, and AI model selection.

Phase 3: Implementation (40-50%)

Goal: Develop the core components of the AI-assisted message router.^[1]_{SEP}**Tasks:**

- Set up a basic pub/sub broker (e.g., configure MQTT, Kafka, or RabbitMQ).
- Implement AI-based routing logic, either via LLM-based message classification (API-based, if feasible) or rule-based AI filtering.
- Integrate message priority and topic-based filtering into the broker.
- Implement a basic monitoring system to track message flow.

Deliverable: A working prototype of the AI-assisted routing system.

Phase 4: Testing & Evaluation (20-25%)

Goal: Assess the performance of the AI-driven message routing system. ^[11]_{SEP}Tasks:

- Design test cases for different message workloads.
- Measure latency, throughput, and routing accuracy.
- Compare AI-assisted routing vs. traditional pub/sub routing.
- Document observations and improvements.

Deliverable: Evaluation report with performance metrics and analysis.

Phase 5: Documentation & Final Report (10-15%)

Goal: Summarize the findings, challenges, and lessons learned. ^[11]_{SEP}Tasks:

- Write a project report explaining the system, methodology, and results.
- Document how the AI-assisted routing was implemented.
- Suggest future improvements and optimizations.
- (Optional) Prepare a demo for presentation.

Deliverable: Final report + presentation/demo.

Adjustments for a 200-Hour Project

If the project is extended, additional advanced features can be explored:

- Multi-broker testing (e.g., implement in both Kafka and MQTT).
- Experiment with different AI models (e.g., fine-tuned NLP model vs. simple rules).
- Implement dynamic rule adaptation, where routing rules evolve over time based on traffic.
- Visual dashboard to monitor AI-based routing performance in real-time.